Getting Started with Point-of-Care Ultrasound for Regional Anaesthesia

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Abstract

Ultrasound-guided regional anaesthesia has a real learning curve, most of it around probe handling and needle visualisation rather than anatomy alone. This orientation guide collects a few practical habits that help new operators progress safely and efficiently, framed around the internationally recommended core competencies for training in ultrasound-guided regional anaesthesia, for use alongside — not instead of — supervised, hands-on teaching.

Keywords: regional anaesthesia, ultrasound, nerve blocks, training

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Departmental guide

Prepared by the Residents' Academic Committee

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Summary. Ultrasound-guided regional anaesthesia has a real learning curve, most of it around probe handling and needle visualisation rather than anatomy

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alone. This orientation guide collects a few practical habits that help new operators progress safely and efficiently, for use alongside — not instead of — supervised, hands-on teaching.

KEYWORDS · REGIONAL ANAESTHESIA · ULTRASOUND · NERVE BLOCKS · TRAINING

1. Probe selection

- **Linear (high-frequency) probes** — best for superficial structures (nerves within ~4 cm), such as interscalene, supraclavicular, and forearm blocks. Better resolution, less penetration.
- **Curvilinear (low-frequency) probes** — needed for deeper structures, such as the sciatic nerve at the gluteal level or in larger patients.

2. Needle visualisation

- **In-plane technique** keeps the entire needle shaft visible along the probe's long axis — generally the easier technique to learn first, as the needle tip position is continuously seen.
- **Out-of-plane technique** shows the needle only as a bright dot crossing the beam — faster for some experienced operators, but easier to lose track of the tip as a beginner.
- Small, deliberate needle movements — rather than large sweeps — make it far easier to keep the tip in view.

3. Common early pitfalls

1. **Chasing the needle with the probe** instead of keeping the probe still and adjusting the needle trajectory — this usually makes tip tracking worse, not better.
2. **Mistaking the needle shadow for the tip** — the brightest point on screen is not always the true tip; small in-and-out jiggles or a burst of local anaesthetic (hydrolocation) can help confirm true tip position.
3. **Poor gel contact** at the probe edges, distorting the image exactly where the needle is entering.

4. Grounding this in the wider training literature

This orientation guide is meant as a first-days primer, not a substitute for a structured training pathway. The joint ASRA/ESRA committee recommendations describe a full residency- and fellowship-based training pathway for ultrasound-guided regional anaesthesia, including the core technical competencies and the didactic and experiential components expected before independent practice [1] — residents progressing beyond this primer should be familiar with that pathway.

5. Using this guide

This is an orientation aid, not a replacement for supervised practice. Residents should complete ultrasound-guided blocks under direct consultant supervision until independently competent, per departmental training requirements.

Have a guide, protocol summary, or teaching resource to contribute? Email the department to have it added here.

References

[1] B. D. Sites, V. W. Chan, J. M. Neal, R. Weller, T. Grau, Z. J. Koscielniak-Nielsen, G. Ivani, The American Society of Regional Anesthesia and Pain Medicine and the European Society of Regional Anaesthesia and Pain Therapy joint committee recommendations for education and training in ultrasound-guided regional anesthesia, Regional Anesthesia and Pain Medicine 34 (1) (2009) 40–46. doi:10.1097/AAP.0b013e3181926779.